JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON MONDAY 06th APRIL 2026) **TIME : 9:00 AM TO 12:00 NOON**

CHEMISTRY

TEST PAPER WITH SOLUTION

SECTION-A

51. An oxide of iron contains 69.9% iron, its empirical formula, is :

(Given : Molar mass of Fe and O are 56 and 16 g mol^{-1} respectively.)

(1) FeO (2) Fe_2O_3

(3) Fe_3O_4 (4) FeO_3

Sol.

	Fe		Oxygen
Moles :	$\frac{69.9}{56}$		$\frac{30.1}{16}$
Molar Ratio :	1	:	1.5
Molar Ratio :	2	:	3

Answer : **Fe_2O_3**

52. If shortest wavelength of hydrogen atom in Lyman series is x, then longest wavelength in Balmer series of He^+ is :

(1) $\frac{9x}{5}$ (2) $\frac{36x}{5}$

(3) $\frac{x}{4}$ (4) $\frac{5x}{9}$

Sol. $\frac{1}{x} = R \times 1^2 \times \left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$ .....(1)

$\frac{1}{\lambda} = R \times 2^2 \times \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ .....(2)

$\frac{x}{\lambda} = R \times 4 \times \frac{5}{4 \times 9}$

$\frac{1}{\lambda} = \frac{1}{x} \times \frac{5}{9}$

$\lambda = \frac{9x}{5}$

53. Match the **List-I** with **List-II** :

List-I Orbital		List-II Radial nodes and nodal plane	
A.	2s	I.	1 Radial node + two nodal planes
B.	3s	II.	1 Radial node + one nodal plane
C.	3p	III.	2 Radial nodes + No nodal plane
D.	4d	IV.	1 Radial node + No nodal plane.

Choose the **correct** answer from the options given below :

(1) A-IV, B-I, C-III, D-II

(2) A-IV, B-II, C-III, D-I

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-III, C-II, D-I

Sol. No. of Radial nodes = $n - \ell - 1$

No. of Angular nodes or nodal planes = ℓ

2s $\Rightarrow$ n = 2, ℓ = 0

1 radial node + zero nodal plane

3s $\Rightarrow$ n = 3, ℓ = 0

2 radial node + zero nodal plane

3p $\Rightarrow$ n = 3, ℓ = 1

1 radial node + 1 nodal plane

4d $\Rightarrow$ n = 4, ℓ = 2

1 radial node + 2 nodal plane

54. The pairs among

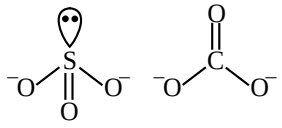
A = [SO_3^{2-}, CO_3^{2-}], B = [O_2^{2-}, F_2], C = [CN^-, CO], D = [NH_3, H_3O^+] and E = [MnO_4^{2-}, CrO_4^{2-}] that do not have similar Lewis dot structure are :

(1) A, B and E (2) A and E

(3) B, C and D (4) C and D

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Sol. A and E do not have similar Lewis structure



MnO_4^{2-} have are unpaired e but CrO_4^{2-} does not have unpaired e.

In B, C, D pair of species haved same no of e.
They have similar Lewis dot structure.

55. Arrange the following isothermal processes in order of the magnitude of the work (p – V) involved between states 1 and 2.

A. Expansion in single stage w_A

B. Expansion in multi stages w_B

C. Compression in single stage w_C

D. Compression in multi stages w_D

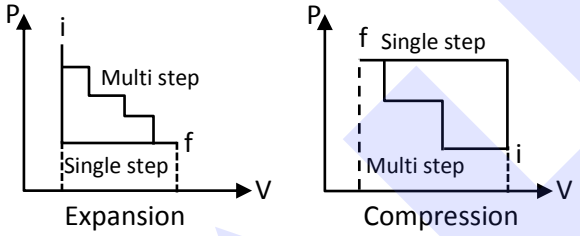
(1) $|w_B| > |w_A| > |w_C| > |w_D|$

(2) $|w_C| > |w_D| > |w_A| > |w_B|$

(3) $|w_C| > |w_D| > |w_B| > |w_A|$

(4) $|w_B| > |w_A| > |w_D| > |w_C|$

Sol.



$\therefore |w_A| < |w_B| < |w_D| < |w_C|$

56. When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was x% of vapor pressure of pure solvent. What is x% ?

(1) 50% (2) 60%

(3) 70% (4) 80%

Sol. $P_S = P^0.X_A$

$$\frac{P_S}{P^0} = X_A = \frac{1}{1+0.25} = 0.80$$

$\% = 0.8 \times 100 = 80$

57. One mole each of He and A(g) are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$A(g) \rightleftharpoons B(g)$

K_C for this reaction at 400 K is 4.0. The partial pressures (in atm) of He and B(g) are respectively (at equilibrium).

(Assume He, A(g) and B(g) behave as ideal gases)

(Given : R = 0.082 L atm K^{-1} mol^{-1})

(1) 3.28, 2.624 (2) 2.624, 3.28

(3) 3.28, 0.656 (4) 0.656, 6.56

Sol.

	A	$\rightleftharpoons$	B
t = 0	1 mol		–
t_{eq^m}	1 – x		x

$$K_C = \frac{x/10}{\frac{1-x}{10}} = 4$$

$x = 0.8$

$$P_{He} = \frac{1 \times 0.082 \times 400}{10} = 3.28 \text{ atm}$$

$$P_B = \frac{0.8 \times 0.082 \times 400}{10} = 2.624 \text{ atm}$$

58. Consider the following data :

Electrolyte	Λ_m^o **(S cm² mol⁻¹)**
$BaCl_2$	x_1
H_2SO_4	x_2
HCl	x_3

$BaSO_4$ is sparingly soluble in water. If the conductivity of the saturated $BaSO_4$ solution is x S cm^{-1} then the solubility product of $BaSO_4$ can be given as.

(Here $\wedge_m = \wedge_m^o$)

(1) $\frac{10^6 x^2}{\alpha^2 (x_1 + x_2 - 2x_3)^2}$ (2) $\frac{x^2}{(x_1 + x_2 - 2x_3)^2}$

(3) $\frac{\alpha^2 (x_1 + x_2 - 2x_3)^2}{10^6 x^2}$ (4) $\frac{x^2}{(x_1 + x_2 + 2x_3)^2}$

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Sol. $BaSO_4(s) \rightleftharpoons Ba^{+2}_{(aq)} + SO^{2-}_{4(aq)}$

s s

$K_{sp} = S^2$

k of $BaSO_4 = x\ S\ cm^{-1}$

Λ^o_m of $BaSO_4 = (x_1 + x_2 - 2x_3)$

$$\Lambda_m = \alpha\Lambda^o_M = \frac{k \times 1000}{S}$$

$$S = \frac{x \times 1000}{\alpha(x_1 + x_2 - 2x_3)}$$

$$k_{sp} = S^2 = \frac{1}{\alpha^2}\left[\frac{x \times 1000}{x_1 + x_2 - 2x_3}\right]^2$$

59. Given below are two statements :

Statement-I : Aluminium is more electropositive than thallium as the standard electrode potential value of $E^o_{Al^{3+}/Al}$ is negative and $E^o_{Tl^{3+}/Tl}$ is positive.

Statement-II : The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms Al^{3+} ion.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

Sol. $E^o_{Al^{3+}/Al} = -1.66\ V$

$E^o_{Tl^{3+}/Tl} = +1.26\ V$

For B : $IE_1 + IE_2 + IE_3 = 801 + 2427 + 3659 = 6887\ kJ$

Both statements are true

60. The **correct** statements among the following are :

A. Basic vanadium oxide is used in the manufacture of H_2SO_4.

B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM.

C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.

D. The number of electrons present in the outer most 'd' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below :

(1) A and B only (2) A, C and D only

(3) C and D only (4) B, C and D only

Sol. A. V_2O_5 is amphoteric. It is used in manufacture of H_2SO_4 .

B. Ziegler natta catalyst is $\underset{+4}{\underline{Ti}}Cl_4 + \underset{+3}{\underline{Al}}Me_3$

$Ti^{4+} : 3d^0\ 4s^0\ n = 0\ \mu = 0$

C. O.S. of Al is +3

D. $\underset{+2}{\underline{Pd}}Cl_2$ is used in wacker process

$Pd^{2+} : 4d^8 5s^0$

C and D are correct.

61. Match the **List-I** with **List-II**

List-I Electronic configuration of tetrahedral metal ion		**List-II Crystal Field Stabilization Energy (Δ_t)**	
A.	d^2	I.	–0.6
B.	d^4	II.	–0.8
C.	d^6	III.	–1.2
D.	d^8	IV.	–0.4

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

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Sol. In tetrahedral complex

$$CFSE = \left[-\frac{3}{5}n_1 + \frac{2}{5}n_2\right]\Delta_t$$

$d^2 : \underbrace{e^{1,1}}_{n_1=2}\ \underbrace{t_2^{0,0,0}}_{n_2=0}$; $C.F.S.E = \left[\frac{-3}{5}(2)\right]\Delta t = -1.2\Delta t$

$d^4 : \underbrace{e^{1,1}}_{n_1=2}\ \underbrace{t_2^{1,1,0}}_{n_2=2}$; $C.F.S.E = \left[\frac{-3}{5}(2)+\frac{2}{5}(2)\right]\Delta t = -0.4\ \Delta t$

$d^6 : \underbrace{e^{2,1}}_{n_1=3}\ \underbrace{t_2^{1,1,1}}_{n_2=3}$; $C.F.S.E = \left[\frac{-3}{5}(3)+\frac{2}{5}(3)\right]\Delta t = -0.6\ \Delta t$

$d^8 : \underbrace{e^{2,2}}_{n_1=4}\ \underbrace{t_2^{2,1,1}}_{n_2=4}$; $C.F.S.E = \left[\frac{-3}{5}(4)+\frac{2}{5}(4)\right]\Delta t = -0.8\ \Delta t$

A - III,B - IV, C - I, D – II

62. Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex ?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$

B. $d_{xy} = d_{yz} > d_{z^2}$

C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$

D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the **correct** answer from the options given below :

(1) A, B and D only (2) A and B only

(3) B and D only (4) B, C and D only

Sol. Order of energy in tetrahedral splitting

$(dxy = dyz = dzx) < d_{x^2-y^2} = d_{z^2}$

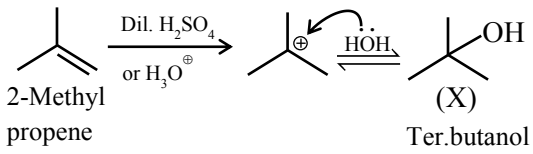
A, B and D are correct.

63. R_f value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42. 2-methylpropene is treated with dilute H_2SO_4 to give major organic product (X). R_f value for (X) in the same solvent system under identical condition will be :

(1) 0.42 (2) 0.82

(3) 0.62 (4) 0.12

Sol.



2-Methyl propene (X) Ter.butanol

Ter butanol (X) is more polar than 2-Methyl propene and $R_f \propto \frac{1}{\text{polarity}}$ so R_f of ter.butanol is less than 2-Methylpropen

So. Ans is 0.12

64. Given below are two statements :

Statement-I : 2, 6-diethylcyclohexanone and 6-methyl-2-n-propylcyclohexanone are metamers.

Statement-II : 2, 2, 6, 6-tetramethylcyclohexanone exhibits keto-enol tautomerism.

In the light of the above statements, choose the **correct** answer from the options given below.

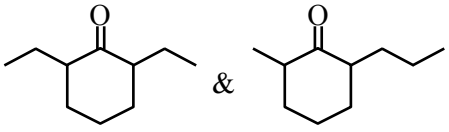
(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

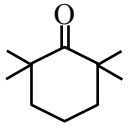
(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

Sol. Statement-I is correct and Statement-II is incorrect



are metamers



No α-H

No Keto enol tautomerism

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65. Given below are two statements :

Statement-I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of CH_3MgBr with water.

Statement-II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

Sol. Methane can not be prepared by Kolbe's electrolysis process. So, statement-1 is false

$$2CH_3-\overset{O}{\overset{\|}{C}}-ONa \xrightarrow{\text{Kolbe electrolysis}} \underset{\text{Ethane}}{CH_3-CH_3} + 2CO_2 + 2NaOH + H_2$$

Methane also can not be prepared by Wurtz reaction so statement-2 is true

$$2R-X \xrightarrow[\text{Wurtz reaction}]{\text{Na.oxyether}} R-R + 2NaX$$

If R = $-CH_3$ then ethane is formed not methane.

66. Given below are two statements :

Statement-I : 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

Statement-II : Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Statement-I and Statement-II are true.

(2) Both Statement-I and Statement-II are false.

(3) Statement-I is true but Statement-II is false.

(4) Statement-I is false but Statement-II is true.

Sol. $Ph-CH_2-CH=CH_2 \xrightarrow{HBr} Ph-CH_2-\overset{\oplus}{C}H-CH_3$

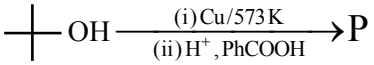
$\downarrow$ Hydride shift

$Ph-\underset{\underset{Br}{|}}{\overset{*}{C}}H-CH_2-CH_3 \xleftarrow{Br^-} Ph-\overset{\oplus}{C}H-CH_2-CH_3$ (Major product)

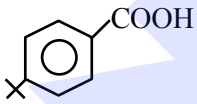
1 Chiral carbon

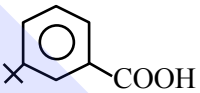
Aryl cyanide is not formed by Gattermann reaction.

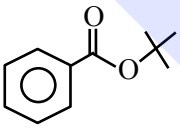
67. Consider the following sequence of reactions

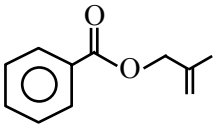


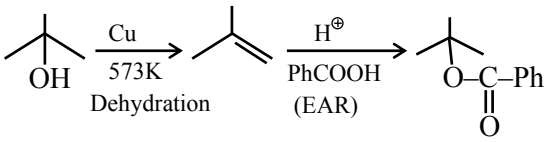
The major product P is :

(1) 

(2) 

(3) 

(4) 

Sol. 

68. Arrange the following compounds according to increasing order of boiling points.

n-C_4H_9OH (A), n-$C_4H_9NH_2$ (B),

n-C_4H_{10} (C) and $C_2H_5NHC_2H_5$(D).

(1) C < B < A < D

(2) D < C < B < A

(3) C < D < B < A

(4) D < B < A < C

Sol. Boiling point $\propto$ Intermolecular attraction

$\propto$ Number of H-bonding

n-butanol has strong H-bonding.

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69. Match the **List-I** with **List-II** :

List-I Deficiency Disease		**List-II Vitamin**	
A.	Scurvy	I.	Pyridoxine
B.	Convulsions	II.	Vitamin A
C.	Cheilosis	III.	Ascorbic Acid
D.	Xerophthalmia	IV.	Riboflavin

Choose the **correct** answer from the options given below :

(1) A-I, B-III, C-II, D-IV

(2) A-I, B-III, C-IV, D-II

(3) A-III, B-I, C-IV, D-II

(4) A-III, B-I, C-II, D-IV

Sol. Vitamin–A → Xerophthalmia

Vitamin-C (Ascorbic acid) → Scurvy

Vitamin-B_2 (Riboflavin) → Cheilosis

Vitamin-B_6 (Pyridoxine) → Convulsions

70. Match the **List-I** with **List-II**

List-I Amino acid		**List-II Positive reaction/Test for functional group present in side chain of amino acid**	
A.	Glutamine	I.	Hinsberg's test
B.	Lysine	II.	Neutral $FeCl_3$ test
C.	Tyrosine	III.	Ceric ammonium nitrate test
D.	Serine	IV.	Hoffman bromamide degradation

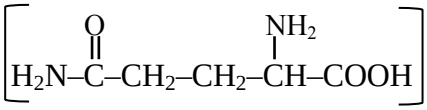
Choose the **correct** answer from the options given below :

(1) A-IV, B-II, C-I, D-III

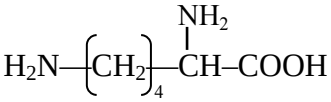
(2) A-IV, B-I, C-II, D-III

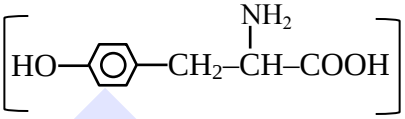
(3) A-III, B-II, C-I, D-IV

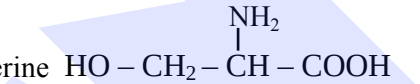
(4) A-IV, B-I, C-III, D-II

Sol. Glutamine 

has amide functional group inside chain. So it gives Hoffmann bromamide reaction.

Lysine  has 1° amine group. So it gives +ve Hinsberg test.

Tyrosine  has phenolic OH. So it gives +ve test with neutral $FeCl_3$.

Serine  has alcohlic –OH group. So it gives +ve test with cerric ammonium nitrate.

SECTION-B

71. First and second ionization enthalpies of lithium are 520 kJ mol^{-1} and 7297 kJ mol^{-1} respectively. Energy required to convert 3.5 mg lithium (g) into $Li^{2+}(g)$ [$Li(g) \rightarrow Li^{2+}(g)$] is _____ kJ mol^{-1}. (nearest integer)

[Molar mass of Li = 7 g mol^{-1}]

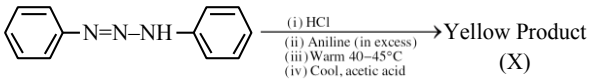
Sol. Number of moles of Li = $\frac{3.5\times10^{-3}}{7} = 5\times10^{-4}$

Total energy needed to convert

$Li(g) \rightarrow Li^{2+}(g)$ is

5×10^{-4} (520 + 7297) = 3.9085 kJ = 4 kJ

72. Consider the following sequence of reactions.



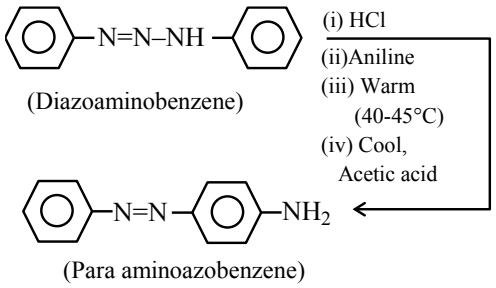
The percentage of nitrogen in the yellow product (X) formed is _____ %.

(Nearest Integer)

(Given Molar mass in g mol^{-1} H : 1, C : 12, N : 14)

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Sol.



- HCl catalysed hydrolysis.
- Aniline acts as a solvent and reactant.
- The mixture is gently heated to facilitate rearrangement.
- After the reaction is complete, the mixture is cooled and treated with acetic acid to neutralise excess acid and precipitate the final product.

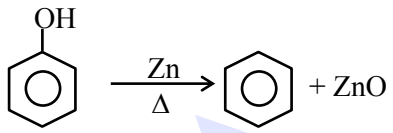
Mass percentage of nitrogen in final product

$= \frac{42}{197} \times 100 = 21.30\ \% \approx 21$

73. 4.7 g of phenol is heated with Zn to give product X. If this reaction goes to 60% completion then the number of moles of compound X formed will be _____ $\times 10^{-2}$ (Nearest Integer)

(Given molar mass in g mol^{-1} : H : 1, C : 12, O : 16)

Sol.



Moles of phenol $= \frac{4.7}{94.0} = 0.05\,mol$

Reaction is 1 : 1 so 0.05 mole of phenol will give 0.05 mole of benzene on 100% yield

For 60% yield = 0.05 × 0.60

= 0.03 mol.

mol of 'X' = 3×10^{-2} mol

So answer is (3)

74. Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2}$ = 3 hour. The percentage of sucrose remaining after 6 hours is ____ . (Nearest integer)

(Given : log 2 = 0.3010 and log 3 = 0.4771)

Sol. Sucrose + $H_2O \rightarrow$ glucose + fructose

$t_{1/2}$ = 3 hr

For 1st order $t_{75\%} = 2 \times t_{1/2}$

So t = 6 hr, 75% of sucrose decomposed

So 25% of sucrose remains after 6 hr.

75. Consider the reaction $X \rightleftharpoons Y$ at 300 K. If ΔH^{θ} and K are 28.40 kJ mol^{-1} and 1.8×10^{-7} at the same temperature, then the magnitude of ΔS^{θ} for the reaction in J K^{-1} mol^{-1} is _____.(Nearest Integer). (Given : R = 8.3 JK^{-1} mol^{-1}, ln10 = 2.3, log 3 = 0.48, log 2 = 0.30

Sol. $X \rightleftharpoons Y$ $K_{eq} = 1.8 \times 10^{-7}$

$$\ln(K_{eq}) = \frac{\Delta_r S^o}{R} - \frac{\Delta_r H^o}{R}\left(\frac{1}{T}\right)$$

$$2.3 \log (1.8 \times 10^{-7}) = \frac{\Delta_r S^o}{R} - \frac{28400}{R} \times \frac{1}{300}$$

$$\Delta_r S^o = 2.3 \times R \log (1.8 \times 10^{-7}) + \frac{284}{3}$$

$$= 19.09\left[-8 + 2 \times 0.48 + 0.3\right] + \frac{284}{3}$$

$$= -128.66 + \frac{284}{3}$$

$$\Delta_r S^o = -33.99 \simeq -34\ J/K.mol$$

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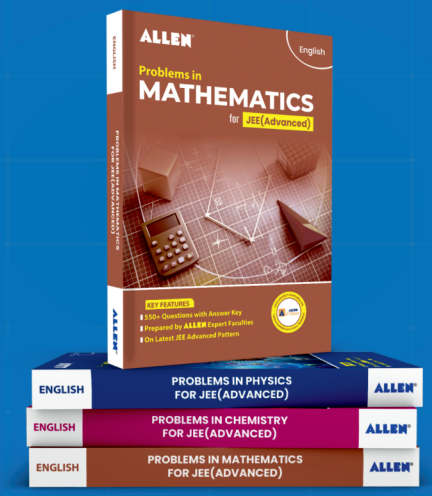


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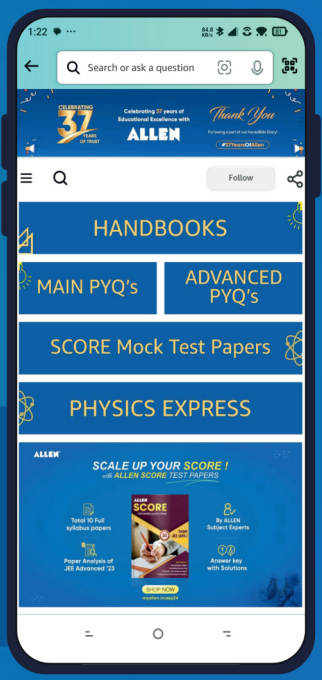
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